Fiscal Year:	FY 2008	**Task Last Updated:**	FY 03/25/2008
PI Name:	Dulchavsky, Scott A. M.D., Ph.D.		
Project Title:	Advanced Diagnostic Ultrasound in Microgravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Web Page:			
City:	Detroit	**State:**	MI
Zip Code:	48202-2608	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	05/01/2003	**End Date:**	12/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS 8, 9, 10, 11, 12 In flight development phase (data collection has begun) NOTE: Received NCE to 12/31/2007 per PI (8/07) NOTE: Received NCE to 07/01/2007 per PI (12/06) NOTE: Moved to Henry Ford Hospital from Wayne St U Hosp early 2006 (12/06)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Sargsyan, Ashot (Wyle Laboratories) Hamilton, Douglas (Wyle Laboratories) Melton, Shannon (Wyle Laboratories)		
Grant/Contract No.:	NNJ04HB07A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The diagnosis of acute health problems in space is problematic due to limited training of the Crew Medical Officer (CMO), human and environmental factors, and changes in anatomy and disease presentation in microgravity. Trauma and other acute clinical conditions may cause serious mission ending health consequences. There is no planned radiological capability aboard the ISS further complicating medical diagnosis in space, however, terrestrial investigations have suggested expanded diagnostic applications of ultrasound in abdominal/thoracic trauma, bony and ligamentous injury, dental/sinus infections, opthalmic, and urologic complications. The Health Research Facility (HRF) will have ultrasound capability for research and possibly operational use. This proposal will optimize diagnostic ultrasound in these applications on the ground, confirm the utility in the Microgravity Research Facility, and operationalize the techniques in the ISS to fulfill the objectives of this proposal: Maximize the sensitivity and specificity of ultrasound in the diagnosis of abdominal injury, pneumothorax, extremity injury, dental/sinus infection, opthalmic injury, and urologic disorders in ground based testing. Verify ultrasound in parabolic microgravity using animal models of potential aerospace medical conditions: optimize data compression/transmission algorithms for space medical use. Obtain baseline data from ISS ultrasound flight experiments in normal crew member volunteers to guide CMO inflight use of ultrasound for research/diagnostic requirements on site and through telemedical support. The unique constraints imposed by the space environment require the development of novel strategies for crew member health problems including the expanded use of ultrasound. Thoracic ultrasound, initially investigated by NASA as an alternative diagnostic modality for pneumothorax and orthopedic injury, has proven accuracy in terrestrial and microgravity applications and may have wide spread impact in acute care. The ultrasonic diagnostic modalities described in this proposal, when optimized, would provide a significant, clinically relevant advance in space medical capabilities with profound Earth-based ramifications in rural or military applications. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although some of the techniques investigated in this proposal are appropriate only for a microgravity environment, the majority of the diagnostic and therapeutic algorithms are readily transferable to terrestrial medicine including rural or military applications. The ultrasonic diagnostic modalities described in this proposal provide a significant, clinically relevant advance in space medical capabilities with profound Earth-based ramifications. Furthermore, training and procedural applications in this proposal are designed to be completed by minimally trained, non-physicians; verification of this novel concept significantly extends medical care capabilities in rural or third world situations.
Task Progress:	The ADUM investigation undertook the complex task of taking ISS-based imaging to a whole new level by creating and validating reliable means for advanced diagnosis using adaptive multimedia tools, as well as a specialized process of real-time discourse. Data acquisition is complete and the majority of the data has been analyzed. The team successfully established the ability of non-medical operators to obtain diagnostic-quality imagery from almost any organ and system of the body, including applications uncommon or even unknown to terrestrial medicine. The hardware specific requirements to expand the operational use of ultrasound in space need to be fully defined. The HRF Ultrasound System is an advanced multipurpose diagnostic imaging tool; if equipped with a reasonable array of ultrasound probes and supported by valid procedures, it will significantly enhance the ability to timely diagnose, stage, and monitor a wide variety of serious conditions. Depending on the mission specifics, the ultrasound capability in space could also be implemented through newly developed, high-fidelity portable ultrasound devices. The outcome of a medical contingency may be changed drastically, and an unnecessary evacuation may be prevented, if clinical decisions are supported by objective diagnostic information. Therefore, since the HRF Ultrasound system is the only diagnostic imaging device currently manifested on ISS, the use of ultrasound in epidemiologically defined clinical conditions should be thoroughly evaluated for its utility in all applicable clinical situations to support the Crew Health Care System / Integrated Medical System of the Station. Directed ground based investigations utilizing ultrasound to diagnose pneumothorax, urinary system obstruction, dental/facial infections, and musculo-skeletal injury have allowed the utility of the ultrasound technique to be maximized for aerospace and ground based applications. The live interactive sessions of ADUM took advantage of the near-real-time (small delay of less than 2 seconds) communication capability of the ISS to guide the astronaut (or cosmonaut) through equipment adjustments and probe position manipulations on the patient's body. The winning combination of instructional and reference tools and other mutually enhancing solutions allowed the acquisition of multiple target views with a reasonably fast tempo; without these tools the same information would be impossible to obtain in a predictable fashion, while any attempt would require substantially longer time. While the video of the live output enabled complete imaging procedures, high-resolution images were saved and downloaded in standardized international format (DICOM-III) using a different satellite transponder. Thus, ADUM delivered a product that enhances the medical support capability during ISS missions, transforming research hardware into an available operational capability. The concepts and technology are equally applicable to lunar missions, where communication delays would be in the same order of magnitude and could still be termed "near-real-time". Most principles of ADUM stand the test for deeper space missions; however neural network-based image processing and additional image databases would need to replace the real-time guidance expert in the image search phase, in order to pre-select only those images that are the best candidates for transmission and diagnostic interpretation. Operational solutions developed for space flight have elicited interest in terrestrial telemedicine community, as the validated capability goes far beyond what was originally envisioned. As the potential uses of ultrasound in a variety of medical contingencies became clear during this investigation, additional ultrasound views were collected to get more observations of the "space normal" conformation of healthy organs in microgravity.
Bibliography Type:	Description: (Last Updated: 07/24/2025)

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